

# The Sturm-Liouville Eigenvalue Problem

## Recovering $\lambda$ from Boundary Conditions Alone

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### SIRPY · Problem 12 — Why $y'' = -\lambda y$ Is More Interesting Than It Looks

An ongoing series. We are building a public library of problems SIRPY has solved — each entry is one problem, the exact input we handed the solver, the exact output it returned, and an honest read of what happened. No slides. No hand-waving. Numbers you can check.

#### A problem with a hole in it

$$y'' = -\lambda y, \quad y(0) = 0, \quad y(1) = 0$$

This is the classical Sturm–Liouville eigenvalue problem — the vibrating string fixed at both ends, the particle in a box, the first separation-of-variables problem in every PDE course.

It also has a hole in it, and the hole is the point of this entry.

Look at what is being asked. The equation is homogeneous. Both boundary conditions are zero. So the function  **$y = 0$  satisfies every requirement, for every value of  $\lambda$** . Nothing in the problem as written excludes it. Hand this to a solver that takes the statement at face value, and the honest response is a shrug: infinitely many  $\lambda$  work, and the “solution” is nothing.

The problem is under-determined, and the resolution is something a mathematician supplies without thinking: you rule out the trivial solution by requiring the eigenfunction to be non-zero — normalizing it somehow — and then discover that only a discrete set of  $\lambda$  admits a non-trivial solution at all. Those are the eigenvalues. For this problem they are

$$\lambda_n = (n\pi/L)^2 = \pi^2, 4\pi^2, 9\pi^2, \dots$$

The question we were interested in was not whether SIRPY could compute  $\pi^2$ . It was whether SIRPY would **notice the hole**.

## What we handed the solver

```
solve_bvp(order=2, f_str="-l*y", x0=0, x1=1,
          left_bc={0: 0.0, 1: "gamma"}, right_bc={0: 0},
          unknown="l")
```

The equation, the two zero boundary conditions, and one declaration: `unknown="l"` — the constant `gamma` appearing in the equation is not given. We supplied no normalization, no guess, and no hint that the answer should be  $\pi^2$ .

## What came back

```
INFO: eigenvalue mode - sirpy added the normalisation y1(0) = 1
      (the homogeneous problem admits  $y \equiv 0$  for every  $l$ ; the
      normalisation pins the eigenfunction scale, not the eigenvalue).
      Found  $l = 9.869604401089358 \sim (1.0000\pi/L)^2$ .
      Other branches: pass guess=[state..., (n*pi/1)**2] with n = 2, 3, ...
```

Read that first parenthesis. SIRPY identified the exact defect in the problem — *the homogeneous problem admits  $y \equiv 0$  for every  $\lambda$*  — and repaired it by adding the normalization  $y'(0) = 1$ .

Then it said the thing that matters most, and that a careless implementation would get wrong: the normalization **pins the eigenfunction scale, not the eigenvalue**. Choosing  $y'(0) = 1$  fixes how tall the eigenfunction is; it does not influence which  $\lambda$  are admissible. That distinction is the mathematical content of the eigenvalue problem, stated plainly, unprompted, by the solver.

## The number

$$\lambda_1 = 9.869604401089358 \pi^2 = 9.869604401089358$$

Not “agrees to twelve figures.” **Identical** — the difference is exactly zero in double precision, to all sixteen digits a 64-bit machine can hold.

SIRPY also reported the result structurally rather than as a bare number:  $\sim (1.0000\pi/L)^2$ . It recognized what kind of object the answer is, not merely its value.

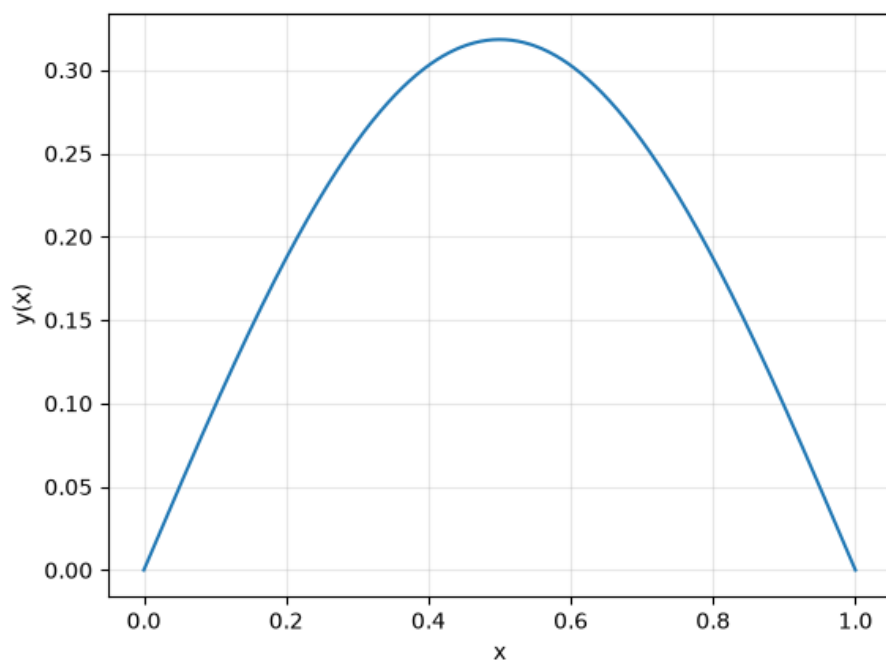
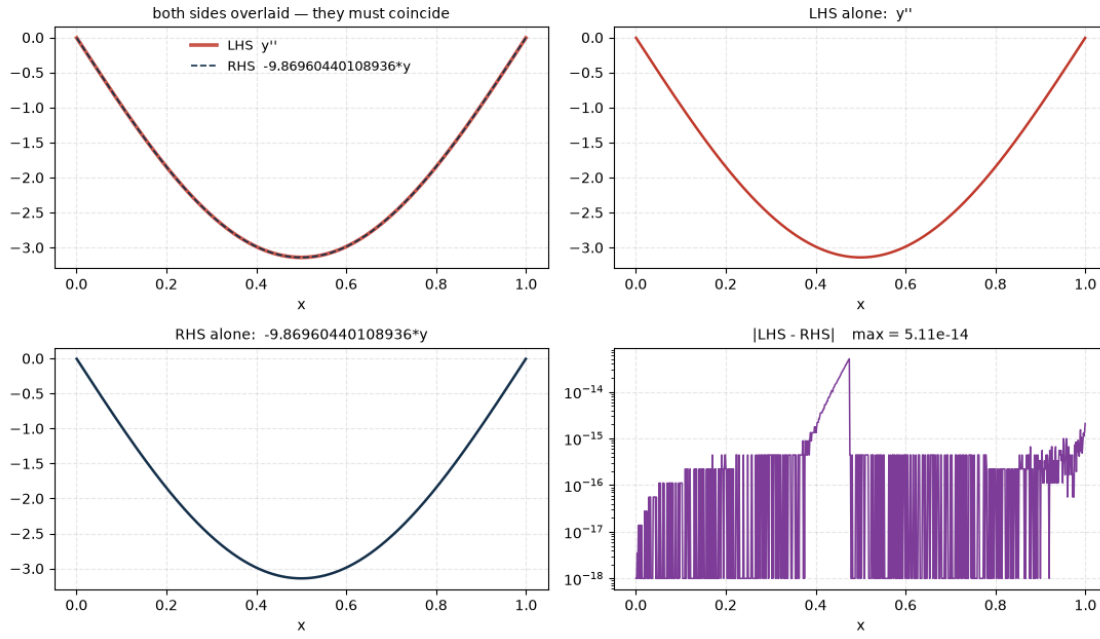


Figure 1: Solution

Visual verification residual  $3.95\text{e-}14$  [VERIFIED]



## The rest of the spectrum

The eigenvalues form an infinite family, and SIRPY said so — including how to reach the others. Following its own instruction:

```
solve_bvp(order=2, f_str="-l*y", x0=0, x1=1,
          left_bc={0: 0.0, 1: "gamma"}, right_bc={0: 0},
          unknown="l", guess=[0, 1, (2*math.pi)**2])
```

Found  $l = 39.47841760435743 \sim (2.0000 \cdot \pi / L)^2$

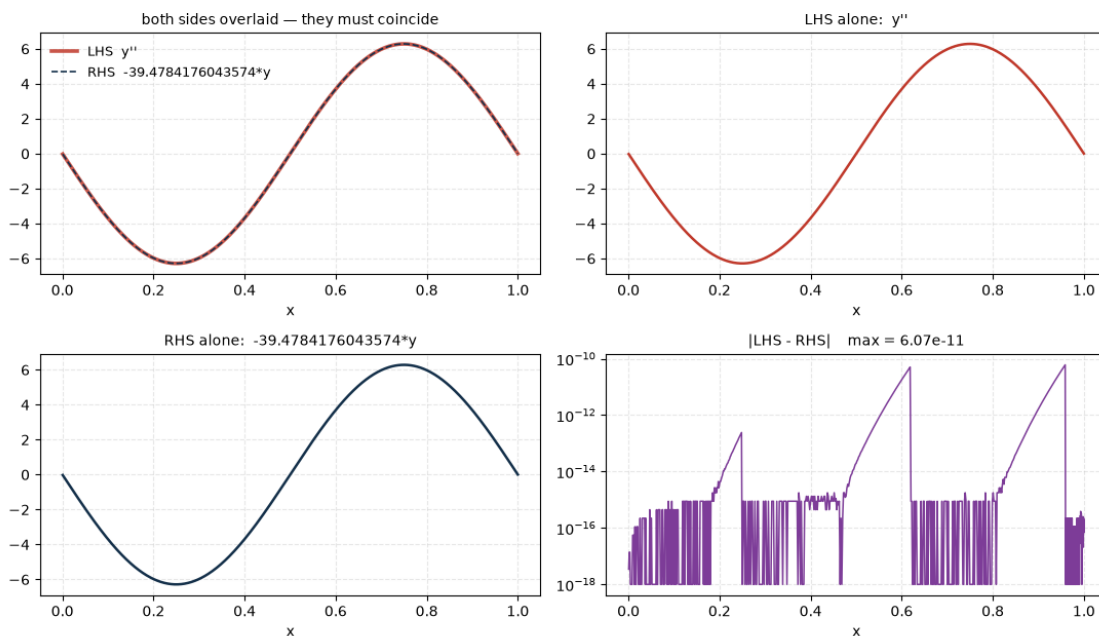
$$\lambda_2 = 39.478417604357432 \cdot 4\pi^2 = 39.478417604357432$$

Identical again, to the last bit.

| mode  | SIRPY              | exact $(n\pi)^2$   | difference |
|-------|--------------------|--------------------|------------|
| n = 1 | 9.869604401089358  | 9.869604401089358  | 0          |
| n = 2 | 39.478417604357432 | 39.478417604357432 | 0          |

The conditions themselves were met to  $5.55 \times 10^{-17}$  and  $1.07 \times 10^{-13}$  respectively.

Visual verification residual 4.74e-11 [VERIFIED]



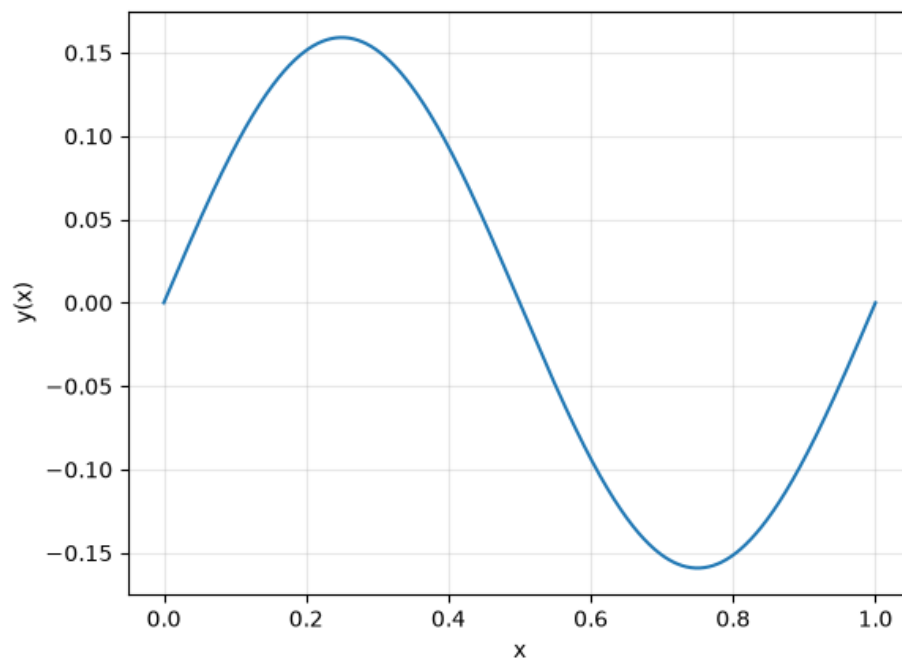


Figure 2: Solution

## What this problem shows

→ **It recognized an under-determined problem** rather than returning the trivial solution or an arbitrary  $\lambda$ . The defect was diagnosed, named, and repaired.

→ **It explained the repair correctly.** The normalization fixes the eigenfunction's scale, not the eigenvalue — the precise statement, volunteered without being asked.

→ **It reported the answer structurally**, as  $(n\pi/L)^2$ , recognizing the form of the result and not just its decimal expansion.

→ **It disclosed that the spectrum is infinite** and gave the exact syntax for reaching higher modes — which then returned  $4\pi^2$  exactly.

→ **Both eigenvalues are exact** to every digit double precision can represent.

## A note on scope

Two modes are not a spectrum. We report exactly what we ran:  $n = 1$  and  $n = 2$ , both exact. Higher modes oscillate more rapidly and are a harder test; we will publish those results as we run them, whatever they show.

## The problem library

This entry is part of the SIRPY Problem Library — an ongoing, public catalog of mathematical systems solved and verified by SIRPY.

- **GitHub** — [github.com/Mitra-Institute-of-Education-MIE/sirpy-problem-library](https://github.com/Mitra-Institute-of-Education-MIE/sirpy-problem-library)
- **Web** — [mitrainstituteofeducation.org/sirpy](https://mitrainstituteofeducation.org/sirpy)

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## Contact Us

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Please only send problems you are free to share publicly.

*SIRPY is in final validation prior to its first public release. Every line quoted above is verbatim from the solver's output; nothing is reproduced from memory.*

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